

BookSprint

Requirements & Specification Document

08/03/2026, Iteration 3

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Yellow headings, backgrounds and **Iteration 3** throughout the document indicate the newest changes.

Purple headings, backgrounds and **Iteration 2** indicate content written during Iteration 2.

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Project Abstract

Booksprint is an application that allows a user to type along a book of their choice. In the typing interface, users type along the book, with it featuring colour coded error correction, the ability to set timers, as well as displaying a user's speed and accuracy. A user can choose a book of their choice from the library (implemented from Google Books API), by either searching for and choosing one directly or selecting a book in the "Discover" area. In-progress books are stored in and can be continued from the "Your Collection" page. They can also review their stats on the "Profile" page. The app provides an engaging and unique way to both improve one's reading and typing simultaneously.

Customer

There are many customers and target users for this application. Examples include children looking to read and learn to type, or teenagers and adults who are looking to read books while honing their typing. Other customers may simply want a way to make reading more stimulating and entertaining, which can include people of all ages.

Competitive Analysis

Typing and reading apps are abundant, but very few applications combine both. Competitors in the same niche include **Entertrained** and **TypeLit**. Here is how our app compares:

- **vs. TypeLit:** BookSprint offers a detailed and sorted "Discover" page that TypeLit lacks
- **vs. Entertrained:** BookSprint features a rotating "Book of the Week," which Entertrained does not have
- **Other Unique Features:** configurable session timers (5 min, 10 min, endless, custom) before typing begins

Personas

Matt Isaacson

Age: 22

Occupation: Student

Tech Comfort:

Role: User



Background:

Matt is a university student who plans to major in literature. As a new student, Matt is finding it stressful to take good notes during lectures and is overwhelmed with the amount of reading. Matt is motivated most by visible progress.

Goals:

Wants to improve typing speed, and find a more engaging way to get through required reading.

Candace Trevor

Age: 58

Occupation: Realtor

Tech Comfort:

Role: Admin



Background:

Candace is a full-time realtor who volunteers what little spare time she has to work as the admin for Booksprint. Candace doesn't have a lot of time to learn complex menus or systems, and isn't the most tech-savvy user.

Goals:

Needs a simple admin interface that allows her to check who's using the system and remove problematic accounts efficiently.

Iteration Progress (After Iteration 2)

The velocity for Iteration 2 resulted in a velocity of 13 story points:

User Story	Owner	Points
Typo Correction	Travis	2
Typing Test with Session Timer	Travis	2
Discover Page and New Books Panel	Ishaan	1
View Collection & Profile	Ishaan	1
User Info Dashboard in Admin View	Max	3
Stream Book Text to Typing Page	Cainan	2
Search & Display Book Metadata	Cainan	1
User Book Collection and Progress	Cainan	1

Velocity: 13 points

The specific improvements that were made for Iteration 3 in order to help increase velocity were:

- Start thinking about and writing User Stories/Tests earlier to help guide development better
- Look ahead to deliverables in general so development pace can remain consistent
- Let everyone know about open Pull Requests to make sure they don't sit open on GitHub for days at a time

The goal for Iteration 3 was to increase velocity to 14, and we succeeded in achieving 16.

Iteration Progress (After Iteration 1)

In the retrospective for Iteration 1, the team assigned story points based on the size and difficulty of the user stories they implemented, which resulted in the following measurements:

Iteration 1 Velocity

User Story	Owner	Points
User Login & Logout (<i>UI implementation</i>)	Ishaan	1
Post-Login Landing Page	Ishaan	1
Typing Practice	Travis	3
New User Registration / User Login & Logout (<i>backend implementations</i>)	Cainan	3
Admin User List View	Max	2

Velocity: 10 points

The team agreed that the main hurdle in Iteration 1 was simply taking too long to start. A more consistent spread of work over the course of Iteration 2 would prevent members from blocking each-other and merge conflicts from happening, both of which in turn should allow for a higher velocity. **The goal for Iteration 2 was to increase velocity to 12, and we succeeded in achieving 13.**

User Stories - Iteration 3

Genre Based Book Recommendations

Iteration 3

Story Points: 2

ACTOR User
ACTION Navigate to the "Discover" page and typing interface and view the "Booksprint Recommendations" and "More like this" area.
RESULT I can discover books based on the genre of my current book.

"As a reader, I want book recommendations based on genres I enjoy so I can find more books that interest me."

Preconditions

- The user is authenticated and past the login page.
- The user has a current active book they are reading.
- The user is on the "Discover" page or on the typing interface.

Postconditions

- The "Discover" page and the "More like this" section below the typing interface displays recommendations based on the user's active book's genre.
- The user can select the books to read them.

Acceptance Tests

1. Genre-based recommendations

Parameters: The user has an active book and navigates to the "Discover" page and the typing interface.

Assertion: Books with the same genre are displayed under "Booksprint Recommendations" and "More like this".

2. Handles limited user data

Parameters: The user does not have an active book to base recommendations off and navigates to the "Discover" page and typing interface.

Assertion: The page remains functional, and the message "Add a book to get recommendations." appears below the "Booksprint Recommendations" and "More like this" title.

3. Recommendations are functional

Parameters: The user has an active book and navigates to the "Discover" page or typing interface and clicks a book play button under "Booksprint recommendations" or "More like this".

Assertion: The book is selected as the new active book and the user is redirected to the typing interface.

View Most Popular Books

Iteration 3

Story Points: 2

ACTOR User
ACTION Navigate to the "Discover" page and view the "Most Popular" section.
RESULT I can discover the most popular books.

"As a reader, I want to view the most popular books to discover new titles other users enjoy."

Preconditions

- The user is authenticated and past the login page.
- The user navigates to the "Discover" page and views the "Most Popular" section.

Postconditions

- The "Most Popular" section displays the most favorited books by users.
- The books are sorted by the amount of favorites.

Acceptance Tests

1. Popular books are displayed

Parameters: The user navigates to the "Discover" page and views the "Most Popular" section.

Assertion: The most favorited books are correctly displayed.

2. Books are displayed based on popularity

Parameters: Multiple books have been favorited by users in differing amounts.

Assertion: The books with the most favorites appear further left on the list.

3. Books are functional

Parameters: The user navigates to the "Discover" page and views the "Most Popular" section, and clicks the play button of a book.

Assertion: The book is selected as the new active book and the user is redirected to the typing interface.

Track Accuracy with a Typing Heatmap

Iteration 3

Story Points: 5

ACTOR	User
ACTION	With any book selected, navigate to the /app page and select or enter a custom time to start a session.
RESULT	The virtual keyboard will track keypresses, show the next key to be typed, and style each key from yellow to red as typos are made.

“As an aspiring touch-typist I want to see a visual representation of my problem keys in a typing session so I can actively work to improve my typing with these keys.”

Preconditions

- User has authenticated by entering their details in at the /login route
- User has selected a book by searching via the "New Book" modal, or by pressing the play button underneath a book in the "Discover" page
- User has selected a session time on /app page

Postconditions

- The next character to be typed will always be indicated by the symbol on the respective key pulsing an amber colour
- For any key where the user makes a mistake, the key will appear yellow and successive errors that degrade the percentage accuracy will tint the key towards a dark red
- Improving the accuracy percentage will reverse the tinting of a key, first towards yellow, and eventually towards normal

Acceptance Tests

1. Heatmap Doesn't Update While Session is Paused

Parameters: A session is in progress, and the user has hit the Escape key, pausing the typing test.

Assertion: Continuing to type with the session paused will not result in the current key being typed in the typing test being tinted towards red.

2. Heatmap is Reset Each Session

Parameters: A session has been completed, either by the timer running out, or by the user ending the session with the End Session button in Endless Mode. At least one key has recorded a typo.

Assertion: Starting a new session on the same book will result in the keyboard being in the normal state with no tinted keys.

Admin Delete User Account

Iteration 3

Story Points: 2

ACTOR Admin

ACTION Admin clicks the "Delete" button next to a user's entry on the /admin page.

RESULT After confirming the action, the user's account and their book collection are permanently removed, and they no longer appear in the admin user list.

"As an admin, I want to delete a user's account so that I can remove spam, test, or unwanted accounts from the platform."

Preconditions

- Admin is authenticated and on the /admin page
- At least one non-admin user account exists

Postconditions

- Admin clicks "Delete" on a user's row and confirms the action
- The target user's account and book collection are deleted, and their row disappears from the list without a page reload

Acceptance Tests

1. Successful Deletion Removes User and Collection

Parameters: Admin clicks Delete on a user who has books in their collection, and confirms the dialog.

Assertion: The user's account and book collection are removed; their row disappears from the admin page.

2. Cannot Delete Own Account

Parameters: Admin clicks Delete on their own row and confirms.

Assertion: The deletion is rejected with a clear error message, and the admin's account remains.

3. Deletion Cancelled

Parameters: Admin clicks Delete but cancels the confirmation dialog.

Assertion: No request is sent, and the user's row remains unchanged.

4. Unauthorized Access

Parameters: A non-admin or unauthenticated request calls the delete endpoint directly.

Assertion: Request is rejected (401/403), and the target account is not deleted.

Admin View of User's Favorite Book

Iteration 3

Story Points: 1

ACTOR Admin

ACTION Admin selects a user on the /admin page and views their book collection.

RESULT The user's favorite book is visually highlighted (starred) within their book list, distinguishing it from the rest of their collection.

“As an admin, I want to see which book a user has marked as their favorite so that I can understand what content resonates most with them.”

Preconditions

- Admin is authenticated and on the /admin page
- At least one user account exists with books in their collection

Postconditions

- Admin selects a user with a favorited book
- That book is marked with a star and distinct styling in the books panel; all other books in the list appear unstyled

Acceptance Tests

1. Favorited Book Is Highlighted

Parameters: Selected users have multiple books in their collection, one marked as favorite.

Assertion: The favorite book's list item is starred and styled distinctly from the others.

2. No Favorite Set

Parameters: Selected users have books in their collection but haven't marked any as favorite.

Assertion: No book in the list is starred or highlighted.

3. No Books At All

Parameters: Selected users have no books in their collection.

Assertion: The books panel shows "No books yet" with no favorite indicator.

4. Unauthorized Access

Parameters: A non-admin or unauthenticated request calls the admin users endpoint directly.

Assertion: Request is rejected (401/403), and no favorite data is returned.

View Personal Typing Analytics

Iteration 3

Story Points: 2

ACTOR User
ACTION Complete one or more typing sessions, then navigate to the "Profile" page.
RESULT I can view statistics from my latest session, my overall performance, and the key I miss most frequently.

"As a user, I want to view my typing statistics on my profile so I can track my performance and identify areas where I can improve."

Preconditions

- The user is authenticated and past the login page.
- The user has completed at least one typing session.
- The user navigates to the "Profile" page.

Postconditions

- The latest session displays the user's WPM, accuracy, words typed, and number of mistakes.
- The overall statistics display the user's personal-best WPM, average WPM, average accuracy, most missed key, number of times that key was missed, and total sessions completed.
- The "Performance Overview" visually displays overall accuracy and compares the latest WPM with the personal-best WPM.
- The statistics persist between visits to the profile page.

Acceptance Tests

1. Latest session statistics are displayed

Parameters: The user completes a typing session and navigates to the "Profile" page.

Assertion: The WPM, accuracy, words typed, and mistakes from the completed session are displayed under "Latest Session".

2. Overall statistics are calculated

Parameters: The user completes multiple typing sessions with different WPM and accuracy results.

Assertion: The profile displays the correct personal-best WPM, average WPM, average accuracy, and number of completed sessions.

3. Most missed key is identified

Parameters: The user completes multiple sessions and makes errors on several different keys.

Assertion: The key with the highest accumulated number of mistakes and the number of times it was missed are displayed under "Overall Statistics".

4. Performance charts reflect the user's statistics

Parameters: The user has completed at least one typing session and navigates to the "Profile" page.

Assertion: The "Overall Accuracy" chart displays the user's average accuracy, and the "WPM Comparison" chart compares the latest WPM with the personal-best WPM.

5. Only completed sessions update overall statistics

Parameters: Typing statistics are temporarily saved while a session is still in progress.

Assertion: The in-progress updates do not increase the completed-session count or affect the user's overall averages until the session is completed.

6. Profile handles users without completed sessions

Parameters: The user has not completed a typing session and navigates to the "Profile" page.

Assertion: The profile remains functional, unavailable statistics display a dash, and completed-session counts display zero.

View Typing Session Summary

Iteration 3

Story Points: 2

ACTOR	User
ACTION	Complete a timed typing session or manually end an Endless Mode session.
RESULT	I can review my session performance and choose to begin another typing session.

"As a user, I want to see a summary after completing a typing session so I can immediately evaluate my performance."

Preconditions

- The user is authenticated and past the login page.
- The user has selected an active book.
- The user has started a timed or Endless Mode typing session.

Postconditions

- A session summary appears when the session ends.
- The summary displays WPM, accuracy, personal-best WPM, mistakes, words typed, correct keystrokes, and the most missed key.
- The completed session's statistics are saved to the user's profile.
- The user can select "Type again" to return to the session-selection interface.

Acceptance Tests

1. Completed session summary is displayed

Parameters: The user completes a timed typing session.

Assertion: A "Session complete" summary appears and displays the session's WPM, accuracy, mistakes, words typed, and correct keystrokes.

2. Personal best is recognized

Parameters: The user completes a session with a WPM higher than their previous personal best.

Assertion: The new WPM is displayed as the personal best and the message "New personal best!" appears.

3. Most missed key is displayed

Parameters: The user makes multiple mistakes on different keys during a session and then completes it.

Assertion: The summary displays the key missed most frequently and the number of times it was missed.

4. Session with no mistakes is handled

Parameters: The user completes a typing session without making a mistake.

Assertion: The session displays zero mistakes and "None" for the most missed key.

5. Endless Mode produces a summary

Parameters: The user starts an Endless Mode session and selects "End Session".

Assertion: The session ends and displays the same analytics as a completed timed session.

6. User can start another session

Parameters: The session summary is visible and the user selects "Type again".

Assertion: The session selection interface appears and allows the user to select a new duration or Endless Mode.

User Stories - Iteration 2

Typo Correction

Iteration 2

Story Points: 2

ACTOR	User
ACTION	An incorrect key is entered during the typing test
RESULT	A subtle wiggle animation is shown for each mistake, and the character's colour is shifted slightly from yellow towards red for each subsequent mistake

“As a motivated user I want to be forced to correct my mistakes so that I can improve my typing accuracy.”

Preconditions

- User is authenticated and on the /app page
- User has started a session by selecting a time or entering a custom value

Postconditions

- User enters an alphanumeric key, a symbol key, or a punctuation key that is different from the character directly to the right of the flashing cursor in the typing interface
- The character to the right of the cursor's colour changes either to yellow, or from yellow towards red, and a wiggle animation plays on the character

Acceptance Tests

1. Input Blocked on Pause

Parameters: Typing test has been started by logging in, selecting a session time, and beginning to type. The ESC key has been pressed

Assertion: No invalid characters typed should cause colour changes or animations to play in the typing interface

Start Typing Test with the Session Timer

Iteration 2

Story Points: 2

ACTOR	User
ACTION	Select a preset time (or endless mode) or enter an integer to specify the duration of the session
RESULT	I can ensure my practice stays consistent without having to track time myself

“As an avid user I want to specify the duration of my typing sessions so I can ensure my practice is consistent and meets my scheduling needs.”

Preconditions

- User has authenticated by entering their details at the /login route

Plus one of the following:

- User has been redirected to the /app route of the site after login
- User has refreshed the page while authenticated on the /app page
- User has manually navigated to /app either from within the site, or manually via typing it in the address bar

Postconditions

- User will select an option for the length of the session via the Session Overlay displayed on top of the typing interface either by clicking one of the preset length buttons (5 min, 10 min, 20 min, or Endless Mode) or by entering an integer value from 1-300 into the text entry field and clicking Start
- The Session Overlay will disappear, and be replaced with the text to be typed, a flashing cursor to indicate the start position, and the Time counter on the top right of the typing interface will be populated with the user's desired session length
- Any subsequent typed character will start the Time, WPM and Accuracy counters

Acceptance Tests

1. Session Overlay Always Appears on Page Load

Parameters: An authenticated user will navigate to /app by logging in and being redirected, or by refreshing the page, or by navigating home via clickable link from all directly linked pages on the site, or by typing the URL in the address bar manually

Assertion: The Session Overlay will appear on top of the typing interface.

2. Typing Input is Blocked Until Session Time is Set

Parameters: With the Session Overlay open, type any lowercase or uppercase letter, any number, the symbols ! @ # \$ % ^ & * () - _ = + [] { } ; : ' " , . < > / ? \ | ` ~ or the ESC key

Assertion: The action will not trigger character highlighting or moving of the cursor on the typing interface below the Session Overlay, and the ESC key will not cause the regular pause actions of dimming the typing overlay or changing the text on the label "Press ESC to Pause"

Find Books Through the Discover Page and New Books Panel

Iteration 2

Story Points: 1

ACTOR User

ACTION Browse books on the "Discover" page and open the "New Book" panel to directly search for books.

RESULT I can explore books and use the interface to search for them.

"As a reader, I want to find new books that interest me to read and add to my collection."

Preconditions

- The user is authenticated and past the login page
- The user clicks either the "Discover" button or "New Book" button

Postconditions

- The "Discover" page displays all book cards and a book of the week
- The "New Book" button opens a panel that allows a user to enter text
- Clicking the X in the top right of the panel closes it

Acceptance Tests

1. "Discover" page displays correct content

Parameters: The user navigates to the "Discover" page.

Assertion: The correct content including book panels, placeholder titles, and play button are displayed.

2. "New Book" panel opens

Parameters: The user clicks the "New Book" button.

Assertion: The panel appears above the page and displays the intended content including the search bar and an "X" in the top-right corner.

3. "New Book" form accepts input

Parameters: The "New Book" panel is open.

Assertion: The user can enter text input in the search bar which is then displayed.

4. "New Book" panel can close

Parameters: The "New Book" panel is open.

Assertion: The user can press the "X" in the top right corner to exit and close the panel.

View Collection and Profile

Iteration 2

Story Points: 1

ACTOR User

ACTION Navigate the "Your Collection" and "Profile" page to view information.

RESULT I can view my information through organized interfaces.

"As a reader, I want to see my collection and profile so I can review my books, progress, and statistics."

Preconditions

- The user is authenticated and past the login page.
- The user clicks either the "Your Collection" button or "Profile" tab under settings.

Postconditions

- The "Your Collection" page displays books a user would have been reading.
- The "Profile" page displays the user's name, role, statistics, and favorite books.

Acceptance Tests

1. "Your Collection" page displays correct content

Parameters: The user navigates to the "Your Collection" page.

Assertion: The correct content including the book panel, progress percentage, and play button.

2. "Profile" page displays correct content

Parameters: The user navigates to the "Profile" page.

Assertion: The correct content including the user's name, words per minute, accuracy, and favorite book(s).

User Info Dashboard in Admin View

Iteration 2

Story Points: 3

ACTOR Admin

ACTION Admin logs into the admin account and views the user list.

RESULT Each user's entry displays their most recent WPM and accuracy, the titles of books in their collection, and their "Member since" and "Last active" dates.

"As an admin, I want to see each user's statistics such as typing performance, book collection, and account activity so that I can understand how users are engaging with the app."

Preconditions

- Admin is authenticated and on the /admin page.
- At least one user account exists.

Postconditions

- Admin views the user list.
- Each user entry shows: latest WPM/accuracy, book titles, and Member since, Last active dates.

Acceptance Tests

1. Active User Shows Full Info

Parameters: A user has saved typing stats and added at least one book to their collection.

Assertion: The admin page shows that user's correct WPM, accuracy, and book titles.

2. New/Inactive User Shows Placeholders

Parameters: A user exists but has never typed or added a book.

Assertion: The admin page shows — for WPM/accuracy, "No books yet" for their collection, and matching Member since/Last active dates.

3. Last Active Updates After Activity

Parameters: A user with existing activity performs a new action (e.g. saves new stats).

Assertion: "Last active" reflects the newer timestamp while "Member since" stays fixed.

4. Unauthorized Access

Parameters: A non-admin or unauthenticated request calls the endpoint directly.

Assertion: Request is rejected (401/403), no data returned.

Stream Book Text to the Typing Page

Iteration 2

Story Points: 2

ACTOR	User
ACTION	Add or select a book and start a typing session on the /app page.
RESULT	The book text is displayed in 20-word chunks, with additional chunks loaded as I type.

“As a user I want my selected book to load in small sections so I can type through it continuously from where I stopped.”

Preconditions

- The user has authenticated through the /login route.
- The user has added a Project Gutenberg book that provides plain-text content.
- The user has selected the book from /collection, has just added it through New Book, or has navigated to /app with a book in their collection.

Postconditions

- The selected book's text will be cleaned to remove Project Gutenberg legal text and a repeated table of contents.
- The frontend will request 20-word chunks from /api/collection/{id}/chunks.
- The frontend will keep additional chunks ready so the user can continue typing without an empty section appearing.
- The first displayed word will be based on the user's saved progress.

Acceptance Tests

1. Book Text Loads in Chunks

Parameters: An authenticated user with a book in their collection navigates to /app?book={id} and starts a typing session.

Assertion: Text from the selected book appears, and each backend response contains no more than 20 words.

2. Additional Text Loads While Typing

Parameters: The user types through all the characters in the currently displayed chunk.

Assertion: The next chunk appears and typing can continue without returning to the beginning of the book.

Search and Display Book Metadata

Iteration 2

Story Points: 1

ACTOR	User
ACTION	Open New Book, enter a title or author, and submit the search.
RESULT	Matching Project Gutenberg books are displayed with their metadata.

“As a user I want to search for books and see their title, author, and cover so I can choose the correct book.”

Preconditions

- The user has authenticated through the /login route.
- The user is on /app, /collection, or /discover and has opened the New Book section.

Postconditions

- The backend will search the Project Gutenberg API catalog.
- Each result will display its title, author, and cover image.
- Books without author information will display "Unknown author."
- The user can select Add to collection from a search result.

Acceptance Tests

1. Search Displays Book Metadata

Parameters: The authenticated user searches for "The Gift of the Magi."

Assertion: Matching results display a title, author, and book cover.

User Book Collection and Progress

Iteration 2

Story Points: 1

ACTOR	User
ACTION	Add a book, type in it, and open My Collection.
RESULT	The collection displays the user's books and saved progress.

"As a user I want my books and typing progress saved so I can return to a book where I stopped."

Preconditions

- The user has authenticated through the /login route.
- The user has found a book through the New Book search.

Postconditions

- Selecting Add to collection will associate the book with the authenticated user.
- The /collection page will display each book's title, author, cover, and whole-number progress percentage.
- Progress will update while typing and be saved after completing a chunk, ending a session, or leaving the page.
- Selecting a book's play button will continue from the last typed word.

Acceptance Tests

1. Added Book Appears in Collection

Parameters: The user searches for a book, selects Add to collection, and moves to /collection.

Assertion: The selected book appears with its title, author, cover, and progress percentage.

2. Mid-Typing Progress Is Saved

Parameters: The user types fewer than 20 words, leaves /app, and opens the same book again.

Assertion: The book resumes from the saved word instead of restarting the chunk.

3. Collection Progress Updates

Parameters: The user types part of a book and then opens /collection.

Assertion: The book displays the updated progress as a whole-number percentage.

User Stories - Iteration 1

New User Registration

PERSONA User
ACTION Create an account with my name, email, and password
RESULT I can access the app under my own identity

“As a new user I want to create an account with my name, email, and password so I can access the app under my own identity.”

User Login & Logout

PERSONA User
ACTION Log in with my email and password, and log out to end my session
RESULT I can access my personal account and end my session when I'm finished

“As a registered user I want to log in with my email and password and log out when I'm finished, so that I can access my account and securely end my session when I'm done.”

Post-Login Landing Page

PERSONA User
ACTION See my overall stats and choose what to practice next
RESULT I have a clear picture of my progress and a starting point for my current session

“As an existing user I want to see my overall stats and choose what to practice next, so that I can have a goal for my session.”

Admin User List View

PERSONA Admin
ACTION View a list of all registered users
RESULT I can monitor site usage

“As an admin I want to view a list of all registered users so that I can monitor site usage.”

Typing Practice

PERSONA User
ACTION Type out a passage of text and see my progress and mistakes highlighted as I go
RESULT I can track my progress and improve my typing accuracy

“As a user I want to type a passage of text and see my mistakes highlighted as I go so that I can track my progress and improve my typing accuracy.”

Test Cases - Iteration 3

Test 13: Genre Based Book Recommendations Iteration 3

User story: Genre Based Book Recommendations

Input

- Login and navigate to the typing interface and view "More like this"
- Navigate to the "Discover" page and view "Booksprint Recommendations"
- Select any book available on the "Discover" page
- Redirected to the typing interface, view "More like this"
- Select a book from "More like this"
- Navigate to the "Discover" page and view "Booksprint Recommendations"
- Select a book from "Booksprint Recommendations"

Tests

- Correct recommendations visible under "More like this" in the typing interface and "Booksprint Recommendations" on the "Discover" page
- When the user does not have an active book underneath those titles, the text "Add a book to get recommendations." is displayed.
- A user can successfully select a book from those areas

Output

- The titles successfully display "Add a book to get recommendations." when a user does not have an active book to base recommendations on
- Once a user has an active book, recommendations are displayed below both titles
- Books are functional and redirect to typing interface to type with once selected

Tests For:

Success

Test 14: View and Test Most Popular Books Iteration 3

User story: View Most Popular Books

Input

- Login and navigate to the "Discover" page
- View "Most Popular" section
- Click the play button on the leftmost book

Tests

- "Most Popular" section displays most favorited books by users correctly
- Books can be accessed

Output

- The section successfully displays the most popular books
- The books are functional and can be accessed

Tests For:

Success

Test 15: Admin Deletes a User Account Iteration 3

User story: Admin Delete User Account

Input

- Register a new user ("Target User") and add a book to their collection
- Log in as admin and navigate to /admin
- Click "Delete" on Target User's row and confirm the dialog
- Click "Delete" on the admin's own row and confirm the dialog
- Click "Delete" on another user's row but cancel the confirmation dialog
- Attempt a DELETE request to the admin users endpoint as a standard (non-admin) user, and while logged out

Tests

- Target User's row disappears and their account/collection are removed from the database after confirming deletion
- Attempting to delete the admin's own account is rejected with a clear error message, and the account still exists
- Cancelling the confirmation dialog leaves the row, account, and collection unchanged
- A standard user or unauthenticated request to the delete endpoint is rejected, and no account is deleted

Output

- Target User: row removed from admin page; account and book collection deleted from the database
- Self-delete attempt: error message "Cannot delete your own account." shown; admin account still exists
- Cancelled deletion: row, account, and collection unchanged; no request sent
- Standard user/unauthenticated request: 403/401 returned; no data deleted

Tests For:

Success

Failure

Test 16: Favorite Book Highlighted in Admin Books Panel Iteration 3

User story: Admin View of User's Favorite Book

Input

- Register a new user ("Fave Test") and add two books to their collection
- Mark one of the two books as favorite via the app's favorite button
- Register a second user ("No Fave Test") and add one book to their collection, without marking any as favorite
- Register a third user ("Empty Test") who adds no books at all
- Log in as admin and navigate to /admin
- Select each of the three users' rows in turn
- Attempt a GET request to the admin users endpoint as a standard (non-admin) user, and while logged out

Tests

- Fave Test's books panel shows the favorited book starred and styled differently from the non-favorited book
- No Fave Test's books panel shows their book listed with no star or highlight
- Empty Test's books panel shows "No books yet" with no favorite indicator
- A standard user or unauthenticated request to the admin users endpoint is rejected

Output

- Fave Test: favorited book shown with a star and distinct styling; other book shown plainly
- No Fave Test: book shown plainly, no star
- Empty Test: "No books yet" message shown, no star
- Standard user/unauthenticated request: 403/401 returned, no data returned

Tests For:

Success

Failure

Test 17: View Personal Typing Analytics Iteration 3

User story: View Personal Typing Analytics

Input

- Register or log in as a user and select a book
- Complete a typing session with a recorded WPM, accuracy, number of words, mistakes, and missed keys
- Navigate to the Profile page and view Latest Session, Overall Statistics, and Performance Overview
- Return to the typing interface and complete another session with different results
- Navigate back to the Profile page
- Log in as a user who has not completed a typing session and navigate to the Profile page

Tests

- The most recently completed session's WPM, accuracy, words typed, and mistakes are displayed under Latest Session
- The personal-best WPM displays the highest WPM achieved across all completed sessions
- The average WPM and average accuracy are calculated from completed sessions
- The total number of completed sessions is displayed correctly
- The most frequently missed key and its accumulated number of mistakes are displayed
- The Overall Accuracy chart reflects the user's average accuracy
- The WPM Comparison chart compares the latest WPM with the personal best WPM; a user without any completed sessions can access the profile without an error

Output

- The latest completed session is displayed correctly under Latest Session
- The personal best, averages, most missed key, missed key count, and completed session count are displayed under Overall Statistics
- The performance charts correctly represent the user's accuracy and WPM
- Statistics persist after leaving and returning to the profile
- Users without completed sessions see dashes for unavailable statistics and zero completed sessions

Tests For:

Success

Failure

Test 18: View Typing Session Summary Iteration 3

User story: View Typing Session Summary

Input

- Log in and select an active book
- Start a timed typing session
- Type several words while intentionally making mistakes on different keys
- Allow the timer to expire
- View the "Session complete" summary
- Select "Type again" and begin another session
- Complete a session with a WPM higher than the previous personal best
- Start an Endless Mode session and select "End Session"
- Complete a session without making any mistakes

Tests

- Completing a timed session opens the session summary
- The summary displays WPM, accuracy, personal best, mistakes, words typed, correct keystrokes, and the most missed key
- The most frequently missed key and the number of times it was missed are correct
- Beating the previous personal best displays "New personal best!"
- Selecting "Type again" returns the user to the session selection interface
- Manually ending an Endless Mode session displays the session summary
- A session without mistakes displays zero mistakes and "None" as the most missed key
- The completed session's statistics are saved and displayed on the user's profile

Output

- The "Session complete" summary appears after timed and Endless Mode sessions
- All session statistics are displayed correctly
- A new personal best is identified and saved when achieved
- The user can return to the session-selection interface by selecting "Type again"
- Sessions without mistakes are handled correctly; the completed session updates the user's profile analytics

Tests For:

Success

Failure

Test 19: Typing Heatmap Tints Keys on Improvements and Mistakes Iteration 3

User story: Track Accuracy with a Typing Heatmap

Input

- Authenticate and wait for redirect to /app route
- Open DevTools (`F12` or `CTRL + I` depending on browser)
- Click the New Book button on the left-hand side of the screen
- Search for "frankenstein" and hit enter
- Press the "Add to Collection" button
- Click the "5 min", "10 min", "20 min", "Endless Mode" buttons in the Session Overlay or entering an integer value between 1-300 in Custom Time field
- Type the first word correctly, then type an incorrect character in place of space

Tests

- Spacebar key shows as red after first error
- Additional error reduces accuracy
- Typing ~15 correct spacebar characters improves the accuracy

Output

- Error displays correctly: Typing an incorrect character in place of space results in the spacebar being light red (not the maximum value)
- Accuracy reduces on successive typo: An additional error changes the key to the darkest possible red with a value of: `rgba(153, 27, 27, 0.55)`
- Successive correct keypresses improve accuracy: The key tints towards yellow at around 10 keypresses, and at exactly 17 the key is restored to its normal state

Tests For:

Success

Test Cases - Iteration 2

Test 4: Session Timer Lifecycle Iteration 2

User story: Start Typing Test with the Session Timer

Input

- Clicking the "5 min", "10 min", "20 min", "Endless Mode" buttons in the Session Overlay or entering an integer value between 1-300 in Custom Time field

Tests

- Timer runs for the correct amount of time
- Input is blocked when timer runs out and Session Overlay opens
- Invalid output in Custom Time field is rejected with message
- User stats are saved when timer runs out

Output

- Timer correctness: All values work as intended
- Input blocking: Input is not propagated beneath Session Overlay
- Invalid Custom Time Input: Displays 'Enter a value between 1 and 300 minutes'
- User Stats: Admin page shows exact values present when timer ran out

Tests For:

Success

Failure

Test 5: Error Dialog Triggered When Continuous Fetch Errors Drain Buffer Iteration 2

User story: Stream Book Text to the Typing Page

Input

- Click the Play button underneath a book on the Collections screen or in the New Book modal dialog
- Clicking the "5 min", "10 min", "20 min", "Endless Mode" buttons in the Session Overlay or entering an integer value between 1-300 in Custom Time field

Tests

- Disable Fetching Text Chunks:
 - Open Browse Dev Tools
 - Open Network Tab
 - Right click text chunk request and select "Block Request URL"
 - Enter the pattern: `/api/collection/\*/chunks\*`
 - Type until all populated chunks are emptied from buffer

Output

- Error modal opens with the message: "We're having trouble loading more of the book. Please check your connection and try again."
- Closing the modal attempts to populate the buffer again and triggers the error again if unsuccessful

Tests For:

Failure

Test 6: Typing Substitutions Accepted For Arbitrary UTF-8 Characters Iteration 2

User story: Typo Correction

Input

- Click the New Book button on the left-hand side of the screen
- Search for “frankenstein” and hit enter
- Press the “Add to Collection” button
- Clicking the “5 min”, “10 min”, “20 min”, “Endless Mode” buttons in the Session Overlay or entering an integer value between 1-300 in Custom Time field

Tests

- Two dashes (“--”) are required to successfully type the em-dash and proceed, no accuracy penalty received
- Typing an incorrect character first followed by a dash triggers the error animation/styling and effects the Accuracy counter
- Typing a proper dash first followed by an incorrect character triggers the error animation/styling and effects the Accuracy counter

Output

- Two dashes successfully highlights the input as correct with no penalty
- Incorrect + correct character triggers animation/styling and reduces accuracy score
- Correct + incorrect character triggers animation/styling and reduces accuracy score

Tests For:

Success Failure

Test 7: Full User Info in Admin View Displayed Correctly Iteration 2

User story: User Info Dashboard in Admin View

Input

- Register a new user and note the registration time
- Log in as that user, navigate to /app, start a session, and type for a few seconds to trigger a stats save
- Click “New Book,” search for a title, and add it to the collection
- Log in as admin and navigate to /admin
- Also attempt to load the admin users list as the standard user (not admin), and while logged out

Tests

- Active User's row shows correct WPM, accuracy, and book title(s), with “Last active” later than “Member since”
- Idle User's row shows — for WPM/accuracy, “No books yet” for their collection, and matching Member since/Last active dates
- A standard (non-admin) user requesting the admin users list is rejected
- An unauthenticated request to the admin users list is rejected

Output

- Active user's row: correct WPM/accuracy, correct book titles, dates with Last active > Member since
- Inactive/new user's row: —/—/“No books yet”/matching dates
- Standard user request: 403 Forbidden, no data returned
- Unauthenticated request: 401 Unauthorized, no data returned

Tests For:

Success Failure

Test 8: Project Gutenberg Book Search and Metadata Iteration 2

User story: Search and Display Book Metadata

Input

- Click the New Book button, search for "Frankenstein" and submit the form
- Search for a title that does not exist
- Block ``/api/books/search*`` and submit another search

Tests

- Matching books display their title, author, and cover
- A search with no matches displays the correct message
- A failed search request displays an error message
- An unauthenticated search request is rejected

Output

- Frankenstein results display a title, author, and cover
- No Results -> Displays "No matching books were found."
- Failed Request -> Displays "Please try the search again."
- Unauthenticated Request -> Returns 401 Unauthorized and redirects the user to login

Tests For:

Success

Failure

Test 9: Book Text Loading and Chunk Continuation Iteration 2

User story: Stream Book Text to the Typing Page

Input

- Click New Book and search for "Frankenstein"
- Click Add to collection, select a session duration from the Session Overlay, and type through multiple sections of the book
- Request a negative chunk number after the initial text has loaded

Tests

- The book begins after its title, table of contents, and Gutenberg legal text
- Each response contains characters representing no more than 20 words
- Additional chunks appear as the user types
- A negative chunk number is rejected; observe what happens after repeated chunk requests are blocked

Output

- Frankenstein begins with the actual first readable section (skipping legal text, ToC)
- Each successful request displays up to 20 words
- Additional text appears before the displayed text runs out
- Requesting an invalid chunk size returns Bad Request 400
- Buffered text remains available, then typing stops when the buffer is empty

Tests For:

Success

Failure

Test 10: User Collection and Saved Book Progress Iteration 2

User story: User Book Collection and Progress

Input

- Search for a book and click Add to collection, navigate to the collection page
- Open the book and type fewer than 20 words
- End the session and then reopen the book
- Attempt to add the same book again

Tests

- The added book appears in the authenticated user's collection
- The collection displays its title, author, cover, and progress percentage
- Mid-chunk progress remains saved after leaving the typing page
- The book resumes at the first word that has not been completed
- Adding the same book does not create a duplicate collection entry
- An unauthenticated collection request is rejected

Output

- The selected book appears with the correct metadata
- The typing and collection pages show a whole-number percentage
- Reopening the book continues from the saved word
- The existing collection entry is returned instead of creating another
- Returns 401 Unauthorized

Tests For:

Success

Failure

Test 11: Discover Books and Open New Book Panel Iteration 2

User story: Discover Page and New Book Panel

Input

- Login and navigate to the front interface and click the "New Book" button to open panel
- Type in search bar
- Exit using the "X" button in the top right corner
- Navigate to the "Discover" page

Tests

- Clicking the "New Book" button opens the panel
- Search bar allows text to be inputted and displayed
- Clicking "X" in the corner closes the panel
- The "Discover" page displays the correct elements and placeholder book cards

Output

- The panel successfully opens when "New Book" button is clicked
- Text can successfully be typed into the search bar and is displayed
- "X" button successfully closes the panel
- Discover page displays the expected placeholder elements

Tests For:

Success

Test 12: View Collection and Profile Iteration 2

User story: View Collection and Profile

Input

- Login and navigate to the front interface
- Click the "Your Collection" button
- Click the "Settings" button
- In the settings dropdown click the "Profile" button

Tests

- "Your Collection" button opens the page displaying user's books
- "Settings" button has a functioning dropdown
- "Profile" button opens the page displaying user's information

Output

- The "Your Collection" button successfully opens when the corresponding button is clicked and displays user's books
- "Settings" button opens a dropdown menu
- "Profile" button successfully opens corresponding page which displays the user's information

Tests For:

Success

Test Cases - Iteration 1

Test 1: Registration and Session Start

User story: User Login & Logout, New User Registration

Input

- Valid name, email, and password

Tests

- Registration succeeds and returns user info
- Session cookie is issued and works on a follow-up request

Output

- 201 Created and user info returned. Authenticated session is now useable

Tests For:

Success

Test 2: Non-Admin Blocked from Admin Route

User story: Admin User List View

Input

- Current user is logged in as a standard user and not an admin

Tests

- User is denied access to the admin endpoint

Output

- 403 Forbidden

Tests For:

Failure

Test 3: Login and Logout Session Lifecycle

User story: User Login & Logout

Input

- Valid login credentials
- Session cookie from that login

Tests

- Login succeeds and starts a session
- Authenticated request succeeds while session is active
- Logout ends the session
- Same session cookie is rejected after logout

Output

- Login: 200 OK
- Authenticated request: 200 OK
- Logout: 204 No Content
- Unauthenticated request: 401 Unauthorized

Tests For:

Success

Failure

UI Mockups

Discover New Books

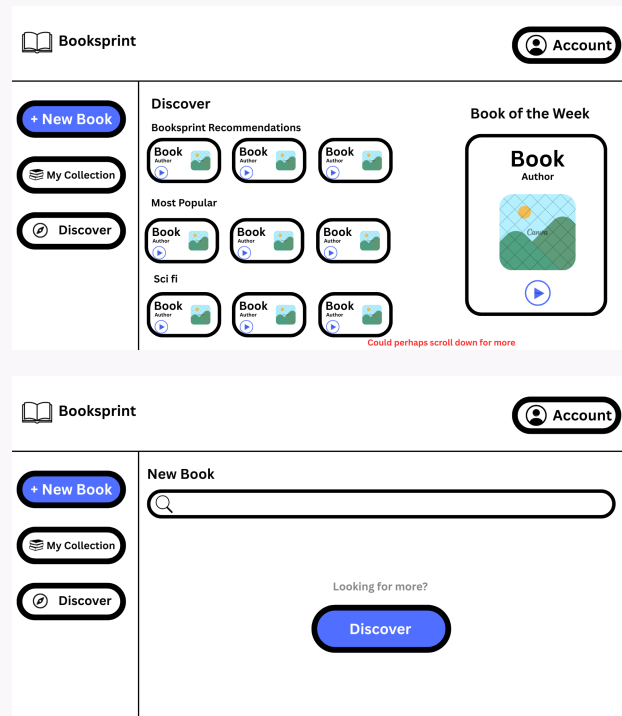
Iteration 2

Overview

Users can search for a specific book, or browse curated recommendations and popular titles if they're not sure what to read and type next.

Key Interactions

- New Book button opens a dialog allowing the user to search for new books without leaving the screen
- Discover page contains recommendations, popular titles, and genre-based lists
- A prominent 'Book of the Week' section



Supports user stories: Find Books Through the Discover Page and New Books Panel

Improved Home Page

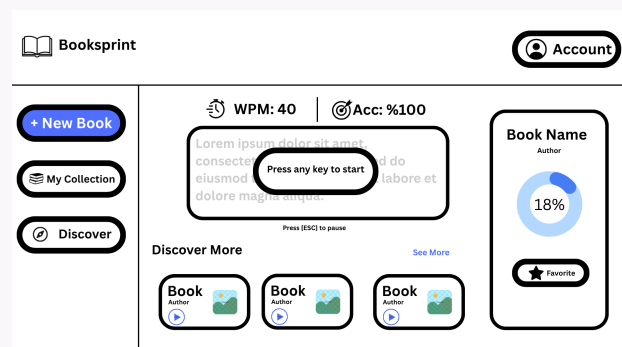
Iteration 2

Overview

The main interface, showing live stats, the text being typed, and quick access to finding more books or viewing the collection.

Key Interactions

- Prominent New Book button allows quick and easy access to starting a new book
- Icons to help elements stand out at a glance
- Live progress toward completing the current book shown as a percentage



Supports user stories: View Collection and Profile, Find Books Through the Discover Page and New Books Panel

Collection & Profile Pages

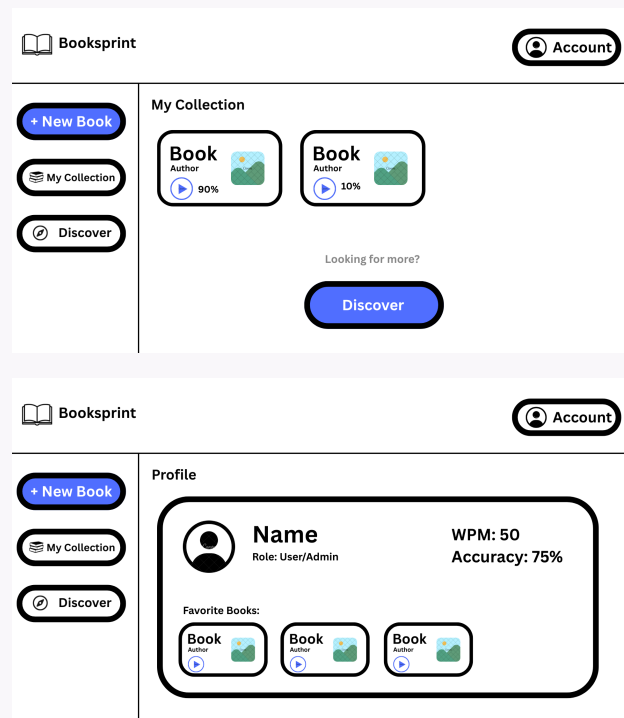
Iteration 2

Overview

Users can view all books they've added along with progress on each, and see an account-level summary of their overall stats and favorite books.

Key Interactions

- Collection view shows each book's completion percentage
- Profile displays aggregate WPM and accuracy ratings
- Favorite books are pinned on the profile page



Supports user stories: View Collection and Profile

Session Overlay

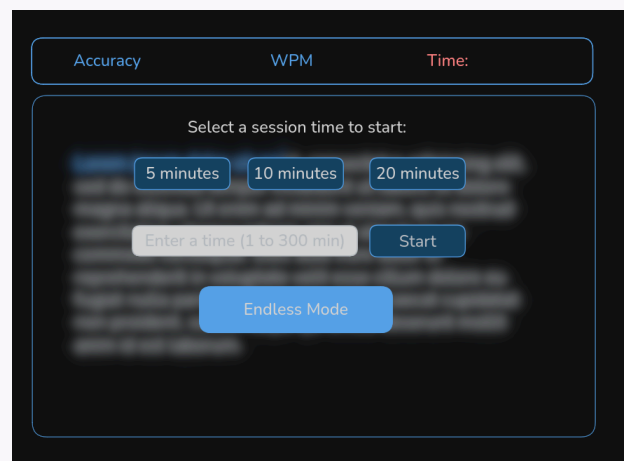
Iteration 2

Overview

Before beginning a typing test allows a user to determine the length of the session.

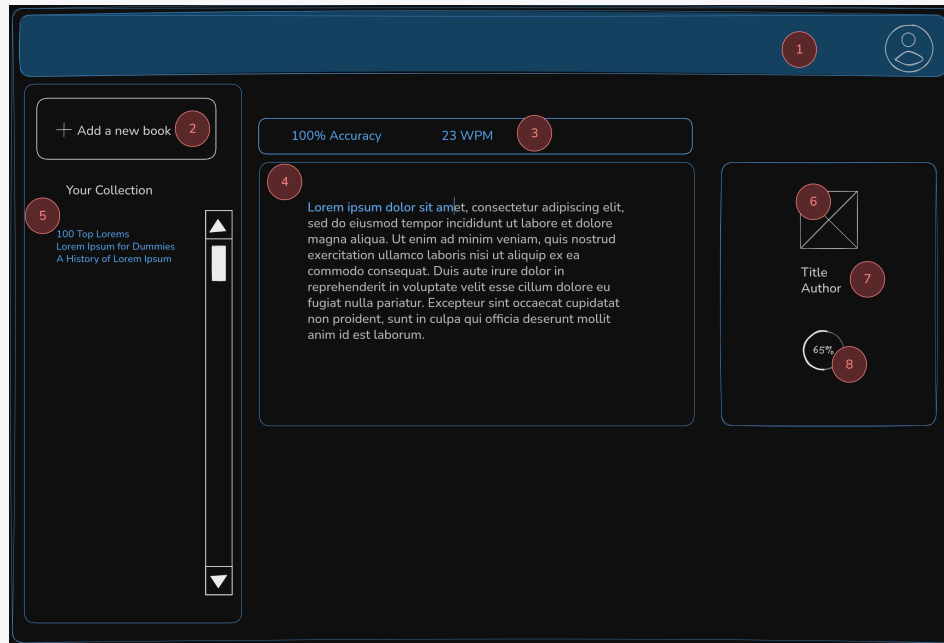
Key Interactions

- Preset options such as 5 minutes, 10 minutes, 20 minutes, or endless to start a session quickly
- Custom time option allows shorter or longer sessions
- Blurred text underneath overlay for a modern look



Supports user stories: Start Typing Test with the Session Timer

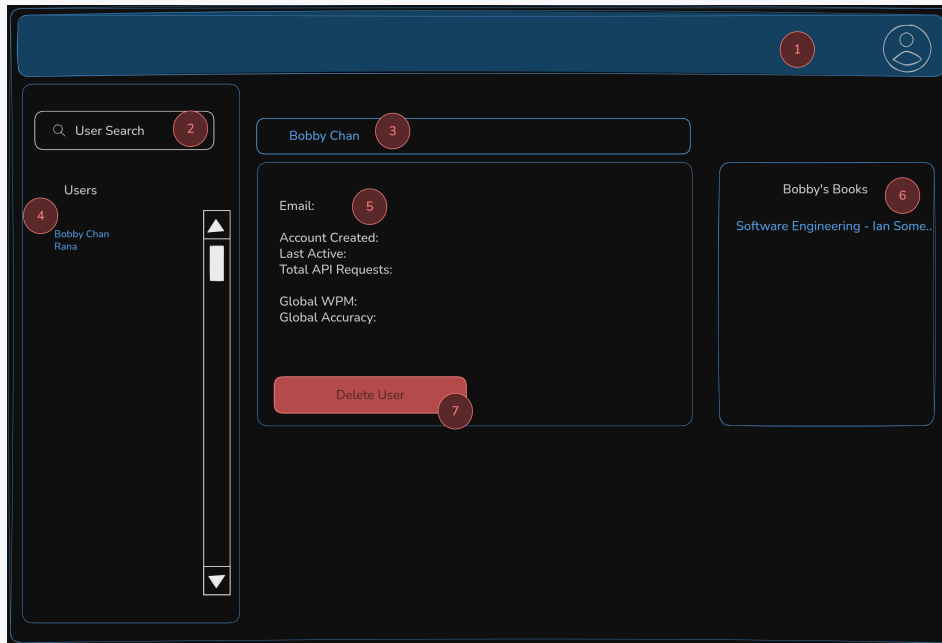
Typing Practice View



1. **Navbar** — Navigation page element present on all pages
2. **Add a New Book** — Opens a modal and allows searching for books present in the Google Books API
3. **Live Stats** — Current live updated stats for the session
4. **Typing Container** — Typed text is emphasized in a darker colour, mistakes in red, and untyped text in light gray
5. **Your Collection** — Changes the typing test/metadata elements to the selected book
6. **Book Cover Image** — Cover image as provided by Google Books API
7. **Book Metadata** — Text metadata about the current book
8. **Progress Indicator** — Show the progress of the user through the current book as a percentage

Supports user stories: Typing Practice

Admin View



1. **Navbar** — Navigation page element present on all pages
2. **User Search** — Filters the User List (Item 4) based on the typed text
3. **Username** — The name attached to the account
4. **User List** — Complete list of users as clickable hyperlinks
5. **User Metadata** — Selected data about the user's account
6. **User Book List** — Lists all books the user has added to their account
7. **Delete User** — Prompts the admin to confirm before removing the account

Supports user stories: Admin user list view